St. George's University
Grenada, West Indies

Basic Principles of Medicine 2

Module: Endocrine and Reproductive System | DLA

# Autonomic Innervation of the Female Reproductive and Pelvic Organs

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# Objectives

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| SOM.MKII.BPM2.1.ER.1.ANAT.1049 | Describe the origin, and type of nerve fibers, found in the superior hypogastric plexus and the inferior hypogastric (pelvic) plexus |
| SOM.MKII.BPM2.1.ER.1.ANAT.1120 | Describe the autonomic innervation of the female pelvic organs and the relationship to the pelvic pain line |
| SOM.MKII.BPM2.1.ER.1.ANAT.1039 | Explain the differences in sensory innervation of the pelvic part of the vagina versus the perineal part of the vagina. |
| SOM.MKII.BPM2.1.ER.1.ANAT.1128 | Compare and contrast the visceral pain pathways for structures above the peritoneal floor of the pelvis versus structures below the peritoneal floor (pelvic pain line) in females |

# Autonomic functions of the pelvic organs

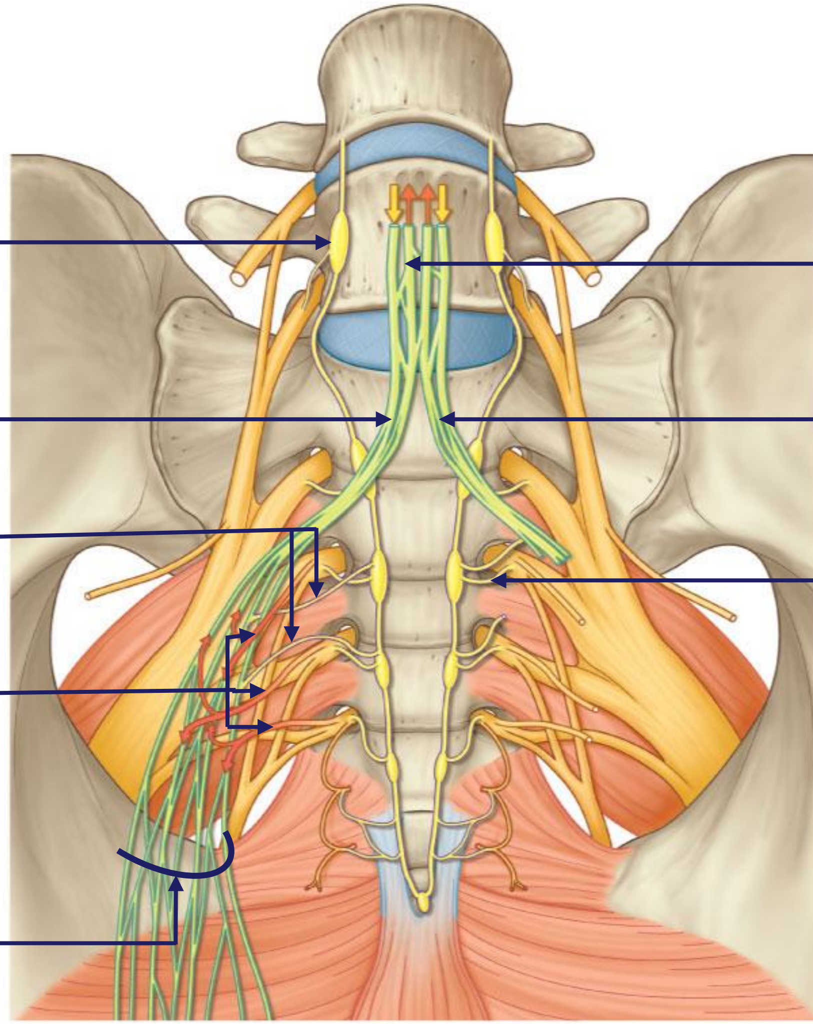
**Sympathetic functions:**

- Contraction of smooth muscle in blood vessels (vasoconstriction)
- Sphincter constriction - internal urethral and internal anal maintaining continence
- Stimulates uterine contraction
- Promotes vaginal and uterine vasoconstriction
- Relaxation of detrusor muscle of the bladder to promotes bladder filling

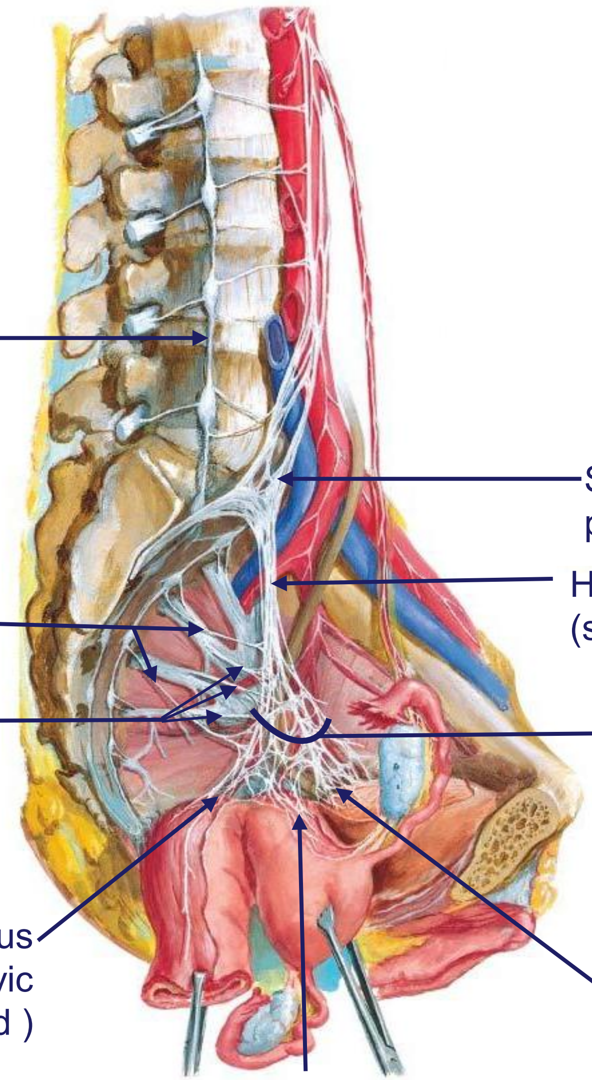
**Parasympathetic functions:**

- Elicits vasodilation of erectile tissues in the clitoris, leading to erection (via the cavernous nerves).
- Stimulates glandular secretions
- Stimulates contraction of the detrusor muscle, promoting micturition
- Sphincter relaxation - internal urethral and internal anal – promoting micturition and defecation
- Stimulates peristalsis in the rectum, aiding in defecation
- Facilitates vaginal lubrication

# Autonomic innervation of pelvic organs



# Autonomic innervation of female pelvic organs



# Autonomic innervation of pelvic organs

- The autonomic nerves of the pelvis form a complex network that is primarily responsible for the regulation of the pelvic organs, including the bladder, rectum, and sexual organs
- These nerves include both sympathetic and parasympathetic fibers, which often have opposing actions

1. **Superior Hypogastric Plexus ( Predominantly Sympathetic)**: This plexus is located anterior to the lower part of the abdominal aorta and bifurcation. It gives rise to the hypogastric nerves

2. **Hypogastric Nerves (Sympathetic)**: These nerves descend from the superior hypogastric plexus along the sides of the rectum to join the pelvic plexuses

# Autonomic innervation of pelvic organs

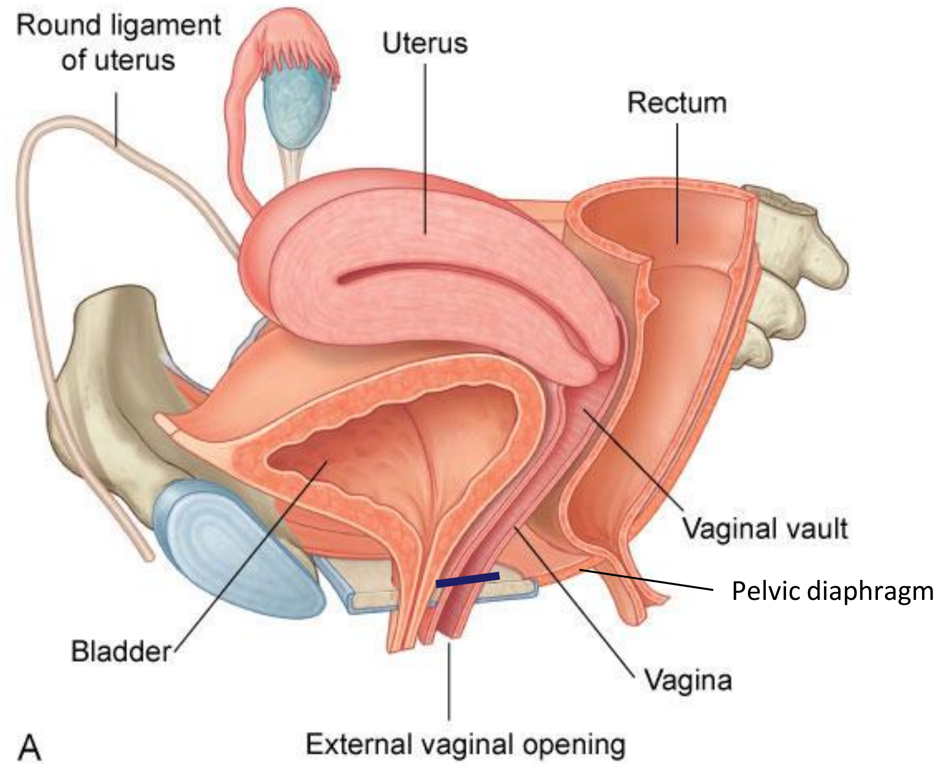
**3. Pelvic Splanchnic Nerves (Parasympathetic)**: These nerves arise from the S2-S4 spinal segments and carry parasympathetic fibers to the pelvic organs

**4. Sacral Splanchnic Nerves (T10-L2) (Sympathetic)**: These nerves arise from the sacral part of the sympathetic trunk, and they carry sympathetic fibers to the pelvic organs

**5. Inferior Hypogastric Plexus (Mixed Sympathetic and Parasympathetic)**: Also known as the pelvic plexus, it lies on the side wall of the pelvis, in front of the piriformis muscle and at the sides of the rectum. **It receives the hypogastric nerves, the sacral splanchnic nerves (sympathetic) and pelvic splanchnic nerves (parasympathetic). Subsidiary plexuses from this plexus innervate the bladder, rectum, sex organs, and other pelvic structures**

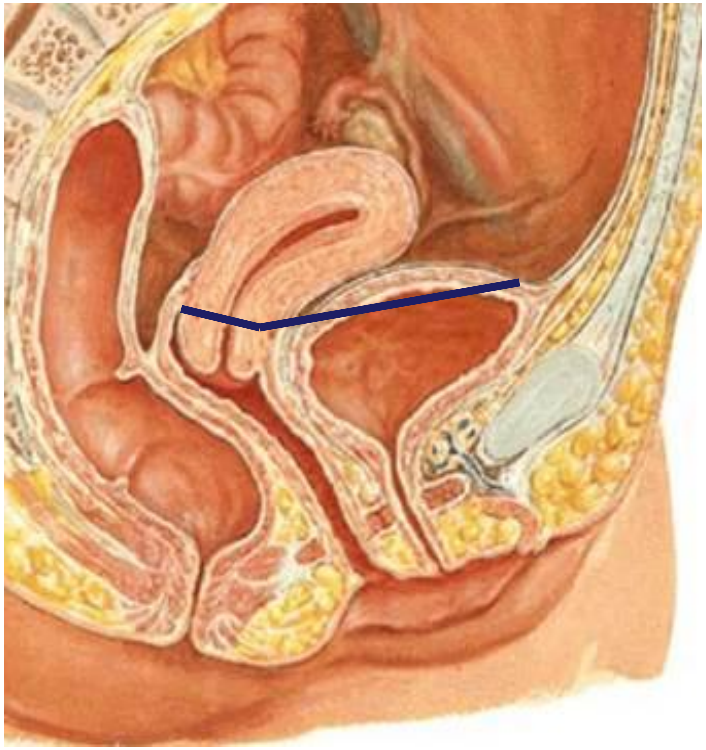
# Autonomic and somatic innervation of the vagina

- The vagina has both autonomic and somatic innervation:
  - The upper two-thirds (pelvic portion) receives autonomic innervation through the uterovaginal nerve plexus (containing sympathetic, parasympathetic and visceral afferent fibers),
  - The lower third (perineal portion) receives somatic innervation via the pudendal nerve, specifically its branch, the deep perineal nerve



# Pelvic pain line (PPL)

- Imaginary line corresponds to the lower limit of the peritoneum
- **Above PPL ( peritoneal part of bladder, uterine tubes, uterine body and fundus)** the visceral afferents for pain travel via the **sympathetics**
    - They travel in the inferior hypogastric plexus
    - Pain is referred to the **T10-L2** areas
- **Below PPL (cervix, majority of urinary bladder, upper 2/3 of vagina)** the visceral afferents for pain travel via the **parasympathetics**
    - They travel in the inferior hypogastric plexus
    - Pain is referred to the **S2-S4** areas



# Pelvic pain line

- The concept of the "pelvic pain line" is often used in anatomy and medicine to understand the patterns of pain sensation in the pelvis. It's essentially a boundary within the pelvis that separates areas with different types of sensory innervation:

  - **Above the Pelvic Pain Line (peritoneal portion of bladder, uterine body, uterine fundus and uterine tubes)**: GVA fibers from structures above the pain line follow the course of the sympathetic efferent fibers. **Pain is referred to the T10-L2 areas**

  - **Below the Pelvic Pain Line (majority of the urinary bladder, cervix and upper 2/3 of vagina)**: GVA fibers from structures below the pain line follow the course of the parasympathetic efferent. **Pain is referred to the S2-S4 areas**